Journal of the Brazilian Society of Mechanical Sciences and Engineering (2022) 44:165
https://doi.org/10.1007/s40430-022-03458-2

TECHNICAL PAPER

Measurement accuracy assessment of the 3D laser triangulation scanner based on the iso-disparity surfaces

Mehdi Irandoust[1] · Sayyed Mohammad Emam[1] · Mohammad Ali Ansari[2]

Received: 8 November 2021 / Accepted: 2 March 2022 / Published online: 3 April 2022

Abstract
The laser triangulation system is one of the most commonly used three-dimensional scanners. The camera resolution is one of the most important parameters that affects the measurement accuracy. Although current high-resolution (HR) cameras have improved the accuracy of scanners, limitations such as high power consumption, high cost, and low signal-to-noise ratio (SNR) have remained. This paper investigates the effect of the camera linear movement (CLM) and the laser rotational movement (LRM) on the measurement accuracy improvement of low resolution/cost laser triangulation scanner. For this purpose, a mathematical model was introduced to calculate the effect of CLM and LRM in the depth reconstruction uncertainty of these scanners. Then, synthetic experiments were performed to evaluate the proposed mathematical model. Finally, real experiments were established to show the advantages of the proposed algorithm to improve the measurement accuracy of the low-resolution (LR) laser triangulation systems practically. The proposed algorithm can be applied in many image processing fields, including reverse engineering, 3D measuring, and 3D modeling. Generally, the presented method is interested in applications that reduce the high instrumental cost and the physical limitations of resolution enhancement through hardware.

Keywords Machine vision · Laser triangulation scanner · Camera linear movement · Laser rotational movement · Quantization error

1 Introduction

In the last decades, machine vision technologies have been developed significantly in measurement and inspection applications [1]. The 3D vision systems can be used for many applications, such as reverse engineering, mapping, creating CAD (Computer-aided design) models, and scanning the ancient artifact for 3D model reconstruction [2–4]. Triangulation-based structured lighting is one of the most common non-contact active dimensional measurement techniques [5, 6]. The main hardware used in these systems is a digital camera and a structured light projector. The triangulation scanners have some limitations generally divided into hardware and software categories. The accuracy of lens, camera sensor resolution, analog-to-digital conversion process, memory usage, and precision of linear and angular stages used to position the camera and laser are the most important hardware parameters affecting the measurement accuracy of laser triangulation scanners [7]. The measurement accuracy can be improved by adjusting the laser scanner setup, such as projected angle, scan depth, and position of the measured part surface to the scanning plane [8, 9]. The camera sensor is one of the most important sources of hardware error affecting the measurement accuracy of laser triangulation scanners. More details can be seen in the images captured by the HR sensors. The cameras with high accuracy and resolution sensor are prohibitively expensive, making them impractical in most applications [10]. Thus, the sensor resolution is limited to represent all scene details. Accordingly, the exact coordinate position of each scene point is approximated by the center coordinate of a pixel and causes the quantization error [11]. The super-resolution algorithms can overcome the limitation mentioned earlier by applying different image processing techniques on the captured images with LR cameras. Changing the relative

Technical Editor: Adriano Almeida Gonçalves Siqueira.

✉ Sayyed Mohammad Emam
sy.m.emam@birjandut.ac.ir

1 Department of Mechanical Engineering, Birjand University of Technology, Birjand, Iran

2 Optical Bio Imaging Laboratory (OBI Lab) Laser and Plasma Research Institute, Shahid Beheshti University, Tehran, Iran

displacement between the LR camera position and the object, known as the dithering technique, is one of the super-resolution methods that can be used to overcome the camera pixel size limitation (image resolution) [12]. The dithering method maintains the image quality acceptable when the number of colors or pixels of an image is reduced during the image processing or when color images convert to gray level in LR printing devices [13]. Another way to improve the image quality of LR display devices is by adding some random noise to the image pixels before it is quantized [14]. The dithering signal (noise) can be applied to the image by moving the camera or object during scanning [12]. The dithering approach can be used in the astronomical camera by different exposures of the same scene with small camera movements [15]. In Laser triangulation scanners, adding the dithering signal for increasing the accuracy of depth reconstruction can be done by changing the position of emitting light by placing and moving a lens in front of the laser [16]. The pattern shifting with the laser scanning rangefinder with the ability to rotate at a certain angular velocity is another method for changing the angular position of the light pattern in sub-pixel order [17]. By applying controlled image motion blur, sampling the space–time volume with higher accuracy to improve the spatial, temporal, and spectral resolution can be achieved [18].

According to the literature review [19–22], the most recent research focused on collecting high-density point clouds using different super-resolution algorithms by LR vision scanners. However, there is less considerable work on the measurement accuracy improvement of LR and low-cost laser systems. For instance, some researchers improved the 3D depth reconstruction of laser triangulation scanners by randomly shifting an object and taking hundreds of LR images [23]. The limitations of this approach are time-consuming, high memory requirement, and the impossibility of controlling the accuracy improvement. It should be noted that there are some types of errors, such as the quantization error, that cannot be reduced even in HR and accurate measurement scanners. In this paper, the measurement accuracy improvement of the LR laser triangulation scanner was investigated. The proposed method is based on analyzing the position and orientation of the iso-disparity surfaces and reducing the quantization error. These surfaces are formed by a special arrangement in front of each laser scanner based on the laser and camera intrinsic and extrinsic parameters. The camera focal length and the relative position and orientation of the laser and camera are the most important parameters to arrange the iso-disparity surfaces. The depth of any object points located exactly on iso-disparity surfaces can be measured without the quantization error. Firstly, the mathematical model of a laser triangulation scanner was presented to calculate the effect of changing the relative position of the camera and laser on the position of iso-disparity surfaces. After that, the synthetic experiments were conducted to validate the mathematical model for improving the measurement accuracy of the laser triangulation scanners with the CLM and LRM methods. Finally, the performance of the proposed algorithm was shown by setting up some physical experiments.

2 The basic principles and calibration of laser triangulation scanner

The camera calibration is used to determine the relationship between the pixel coordinates of an image point and its related metric coordinates [24]. The camera calibration matrix is derived from the intrinsic and extrinsic parameters defined as follows:

1. The extrinsic parameters, including the translation vector and rotation matrix, represent a rigid transformation from the 3D world coordinates to the 3D camera coordinate.
2. The intrinsic parameters represent a projective transformation from the 3D camera coordinate to the 2D image coordinate. These parameters encompass focal length, image sensor format, and principal point.

In this paper, a chessboard that can be easily printed on a rigid surface and visible in different directions was used. Figure 1 shows a schematic representation of the calibration. In this figure, $X_wY_wZ_w$ is the world 3D coordinate reference, $X_cY_cZ_c$ is the camera 3D coordinate reference, and X_dY_d is the image coordinate reference [25].

By using the geometric and trigonometric relationships between the object point and its image in the camera sensor, it is possible to transfer a point in space to the reference of 2D coordinates on the image plane using:

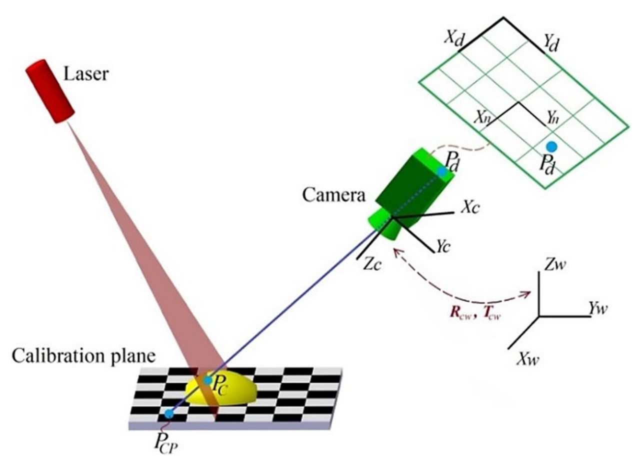


Fig. 1 The schematic of the camera calibration setup

$$s\begin{bmatrix} X_d \\ Y_d \\ 1 \end{bmatrix} = \boldsymbol{A}(\boldsymbol{I}|\boldsymbol{O})\boldsymbol{H}_{cw}\begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}, \quad (1)$$

where s is a random scale factor, and $\boldsymbol{A}$ is the camera intrinsic parameters matrix. This matrix transfers a measured point from the camera 3D coordinate reference to the image 2D coordinate reference. $\boldsymbol{H}_{cw}$ is the camera's extrinsic parameters matrix that transfers from the world coordinate reference to the camera coordinate reference. $(\boldsymbol{I}|\boldsymbol{O})$ is an augmented matrix for transferring coordinates of points from the camera 3D coordinate system to the image 2D coordinate system. In other words, Eq. 1 can be divided into two main equations. In the first one, the coordinate of a point is transferred from the world coordinate reference to the camera coordinate reference according to:

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_{cw} & \boldsymbol{T}_{cw} \\ \boldsymbol{O}_T & 1 \end{bmatrix}\begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}, \quad (2)$$

where $\boldsymbol{R}_{cw}$ is the rotation matrix, $\boldsymbol{T}_{cw}$ is the transition vector, and $\boldsymbol{O}_T$ is the zero matrix to convert the 4×3 matrix to the 4×4 matrix to multiply in a homogeneous coordinate system [25]. In the second one, the three-dimensional point coordinates in the camera coordinate reference are transferred to the two-dimensional coordinate reference of the image using:

$$s\begin{bmatrix} X_d \\ Y_d \\ 1 \end{bmatrix} = \boldsymbol{A}\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}, \quad (3)$$

where

$$\boldsymbol{A} = \begin{bmatrix} \alpha_x & c & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (4)$$

α_x and α_y are the scale factors on the X and Y image axes, respectively. u_0 and v_0 represent the coordinates of the principal point relative to the center of the image frame (X_n,Y_n), and c corresponds to the skewness of the two image axes. Figure 1 shows a schematic image of the laser plane considering the camera and world coordinate systems. An arbitrary point $P_{CP}\left(X_{CP},Y_{CP},Z_{CP}\right)$ can be transferred from the image coordinate reference to the camera coordinate reference using:

$$\begin{bmatrix} X_{CP} \\ Y_{CP} \\ Z_{CP} \end{bmatrix} = s\boldsymbol{A}^{-1}\begin{bmatrix} X_d \\ Y_d \\ 1 \end{bmatrix}, \quad (5)$$

where s is the scale factor of point P_{CP} associated with the camera but on a plane (calibration plane) parallel to the laser plane. The laser plane can be described by:

$$aX_c + bY_c + cZ_c = d, \quad (6)$$

where a, b, c, and d are the constants that describe the laser plane equation. One can find the point $P_C\left(X_C,Y_C,Z_P\right)$ coordinate as the intersection of the laser plane and the $P_{CP}P_d$ line using:

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \frac{d}{aX_{CP} + bY_{CP} + cZ_{CP}}\begin{bmatrix} X_{CP} \\ Y_{CP} \\ Z_{CP} \end{bmatrix}. \quad (7)$$

Z_c is the depth of object point which can be measured by the calibration process.

3 The mathematical model of laser triangulation disparity

According to the sensor resolution limitation, the image coordinate of each scene point is extracted by a pixel center coordinate (the nearest pixel center to the line passes through the lens, scene point, and the camera sensor) that leads to the quantization error. The iso-disparity surfaces can be used to measure the quantization error. These surfaces are positioned in front of stereo vision scanners where the lines that pass through the center of the lens and the pixels' centers of the first camera intersect with the same definition lines of the second camera [26]. Figure 2 shows the iso-disparity surfaces of the standard stereo configuration consisting of two identical cameras with a relative displacement being a pure translation along the cameras' baseline. In this figure, the depth resolution R denotes the distance between each two sequential iso-disparity surfaces. The image of each scene point placed on these surfaces is captured with a pixel center. Therefore, measuring the depth of this point without the quantization error can be possible.

It should be noted that the position and arrangement of the iso-disparity surfaces depend on the optical devices used for the 3D scanners. It is interesting that small changes in the optical devices setups cause important changes to the iso-disparity surfaces [27]. The iso-disparity surfaces for different convergence angles of the stereo vision cameras are shown in Fig. 3.

In the laser triangulation scanner, the position and orientation of the laser and camera (the optical devices) are affected the arrangement of the iso-disparity surfaces. According to our literature review, most researches focused on the iso-disparity surfaces of the stereo vision scanners. However, we could not find any considerable work on calculating the iso-disparity

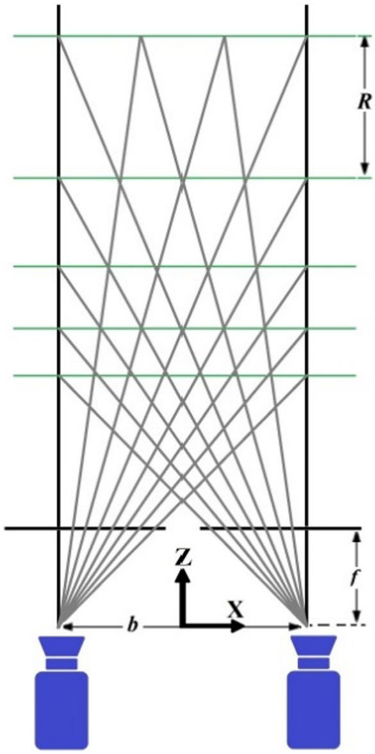


Fig. 2 The stereo vision system disparity model

surfaces of the laser triangulation scanner. The iso-disparity surfaces of the laser scanner are shown in Fig. 4.

The object distance is an important parameter that affects the measurement accuracy. According to Fig. 4, increasing the object distance causes to increase R and thus decreases the measurement accuracy. In the following, the proposed laser scanner mathematical model is investigated. The position of the camera and the laser in the xz plane is shown in Fig. 5.

The origin of the xz-coordinate plane is selected at the middle of the baseline b (the line located between the camera and the laser) and the z-axis is perpendicular to the calibration plane positioned beside the object. As shown in Fig. 5, the image coordinates of point P are located at (x, y). Because of the image resolution limitation, the image coordinates are digitized and quantized by the nearest pixel center (x_Q, y_Q). According to Fig. 5, q_h and q_v are the quantization errors in the horizontal and vertical direction, respectively. The optical axis of the camera is a straight line passing through the center of the lens and the image sensor. The convergence angle of the camera (θ) is the angle between the z axis and the optical axis. The laser angle (α) is denoted as the angle between the z axis and the laser beam. Z can be calculated using the trigonometric formulas of the two right triangles POC and POL:

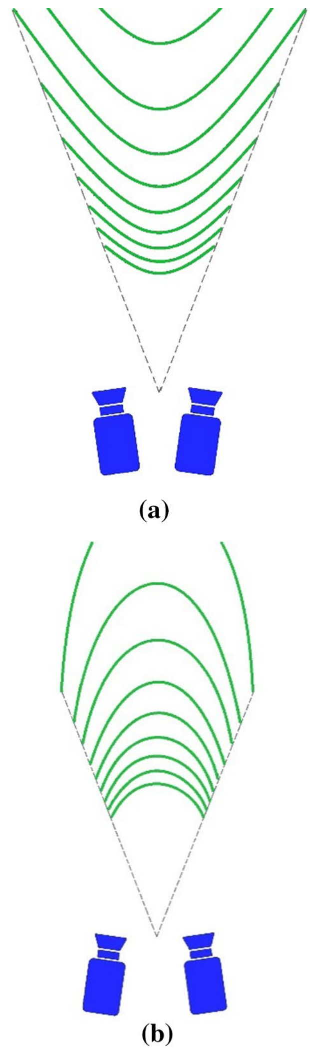


Fig. 3 The iso-disparity surfaces of the stereo vision scanner with the **a** divergence **b** vergence cameras setups

$$\begin{aligned} \mathrm{OL} &= Z \cdot \tan \alpha \\ \mathrm{OC} &= Z \cdot \tan \beta \\ \mathrm{OL} + \mathrm{OC} = b &= Z \cdot (\tan \alpha + \tan \beta). \end{aligned} \tag{8}$$

According to the relationships between the angles α, θ, β, and γ, the depth (Z) of point P can be calculated in terms of convergence angle, laser angle, focal length, and the image coordinate (x) of P according to:

$$\left.\begin{aligned} \tan \beta &= \tan(\theta - \gamma) \\ \tan \gamma &= \tfrac{x}{f} \end{aligned}\right\} \Rightarrow \tan \beta = \frac{\tan \theta - \frac{x}{f}}{1 + \frac{x}{f} \tan \theta}. \tag{9}$$

By substituting Eq. 9 in Eq. 8, the depth of point (P) is calculated by:

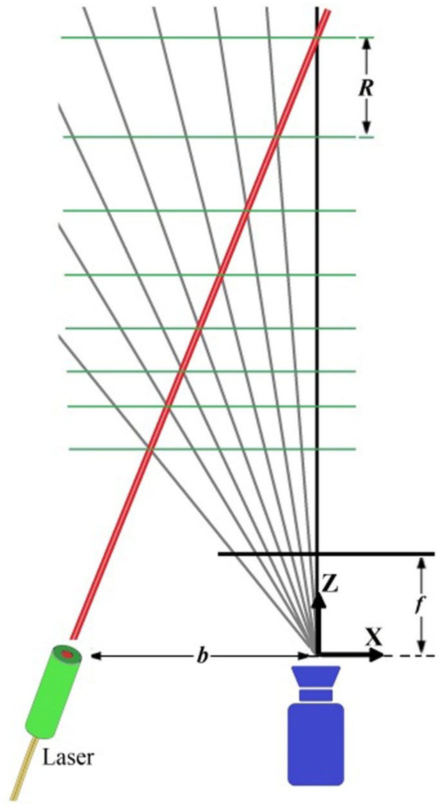


Fig. 4 The laser triangulation system disparity model

$$Z = \frac{b(1 + \frac{x}{f}\tan\theta)}{\tan\alpha + \frac{x}{f}\tan\alpha\tan\theta + \tan\theta - \frac{x}{f}}. \quad (10)$$

According to Fig. 5, the coordinate (x) of P is quantized to the nearest pixel center (x_Q) that can be obtained by $n = \left|\frac{x}{\Delta D}\right|$, where ΔD is the camera's pixel size and | | is rounding off to the nearest integer n. It should be noted that x and ΔD have millimeter units, while n is a dimensionless parameter. Thus, the depth of P after the quantization error (Z_n) can be expressed by:

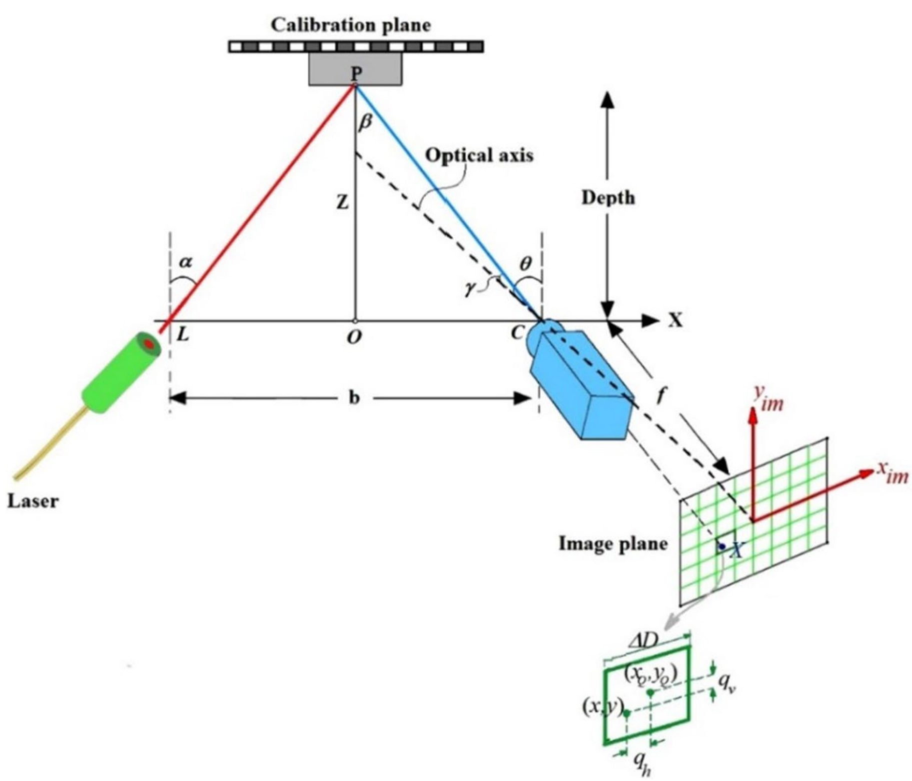
$$Z_n = \frac{b(1 + \frac{n\Delta D}{f}\tan\theta)}{\tan\alpha + \frac{n\Delta D}{f}\tan\alpha\tan\theta + \tan\theta - \frac{n\Delta D}{f}}. \quad (11)$$

Each iso-disparity surface shown in Fig. 4 corresponds to an integer n. The distance between the two consecutive discrete depths Z_n determines the amount of the quantization error, a nonlinear function of n. The distance between two consecutive iso-disparity surfaces ΔZ_n known as the quantization error can be obtained by:

$$\Delta Z_n = \left|Z_{n+1} - Z_n\right|. \quad (12)$$

For 50% improving the measurement accuracy or reducing the quantization error, it will be needed to create a surface between each two consecutive iso-disparity surfaces. For this purpose, new iso-disparity surfaces (Z_n^{new}) should be created with the amount of $\frac{\Delta Z_n}{2}$ relative to its original

Fig. 5 The laser triangulation system geometry model



position. The location of these new surfaces can be calculated using:

$$Z_n^{new} = Z_n + \frac{\Delta Z_n}{2}. \tag{13}$$

According to Eqs. (11–13), the laser angle (α) and the baseline (b) are two important setup parameters that can be changed to create a surface between each two consecutive iso-disparity surfaces. It should be noted that the same findings were achieved using trigonometric similarity relations if the mathematical model was written in the yz plane. The amount of camera linear movement (CLM) and laser rotational movement (LRM) for the presented method is explained in the following sections.

3.1 The mathematical model of CLM

This section presents the mathematical model of using CLM techniques to create a surface between two consecutive iso-disparity surfaces. According to Eq. 14, the new iso-disparity surfaces location Z_n^{new*} can be calculated by only changing b to a new baseline b' in Eq. 11,

$$Z_n^{new*} = \frac{b'(1 + \frac{n\Delta D}{f} \tan\theta)}{\tan\alpha + \frac{n\Delta D}{f}\tan\alpha\tan\theta + \tan\theta - \frac{n\Delta D}{f}}, \tag{14}$$

where b' represents the new baseline required between the camera and the laser to create a surface between two consecutive iso-disparity surfaces. By equating Z_n^{new} with Z_n^{new*}, the new b' can be calculated using:

$$b' = b\left(1 + \frac{\frac{\Delta D}{f}\tan^2\theta}{B}\right), \tag{15}$$

where

$$B = 2(\tan\alpha + \frac{(2n+1)\Delta D}{f}\tan\alpha\tan\theta$$

$$+\left(1 - \frac{n(n+1)\Delta D^2}{f^2}\right)\tan\theta - \frac{(n+1)\Delta D}{f}$$

$$+\frac{n(n+1)\Delta D^2}{f^2}\tan\alpha tan^2\theta + \frac{n\Delta D}{f}\tan^2\theta). \tag{16}$$

It should be noted that both Z_n^{new} and Z_n^{new*} represent the new iso-disparity surfaces and are equal. Therefore, it is possible to change the linear distance between the camera and the laser Δb to create a surface between each two consecutive iso-disparity surfaces using:

$$\Delta b = b' - b. \tag{17}$$

3.2 The mathematical model of LRM

This section presents the mathematical model of creating a surface between two consecutive iso-disparity surfaces using LRM. According to Eq. 18, the new iso-disparity surfaces location Z_n^{new**} can be calculated by only changing α to a new laser angle α' in Eq. 11,

$$Z_n^{new**} = \frac{b(1 + \frac{n\Delta D}{f}\tan\theta)}{\tan\alpha' + \frac{n\Delta D}{f}\tan\alpha'\tan\theta + \tan\theta - \frac{n\Delta D}{f}}, \tag{18}$$

where α' represents the new laser angle required to create a surface between two consecutive iso-disparity surfaces. By equating Z_n^{new} with Z_n^{new**}, the new laser angle α' can be obtained using:

$$\alpha' = \tan^{-1}\left(\frac{b}{Z_n^{new}} + \frac{\frac{n\Delta D}{f} - \tan\theta}{1 + \frac{n\Delta D}{f}\tan\theta}\right). \tag{19}$$

Therefore, it is possible to change the laser angle $\Delta\alpha$ to create a surface between each two consecutive iso-disparity surfaces using:

$$\Delta\alpha = \alpha' - \alpha. \tag{20}$$

4 The synthetic experiment

In this section, two synthetic experiments are shown to demonstrate the performance of the mathematical models. In the first experiment, the effect of CLM and, in the second experiment, the effect of LRM were investigated. In the first experiment, a laser angle (α) of 20°, camera convergence angle (θ) of 30°, and baseline (b) of 250 mm were selected. Figure 6 shows the reconstructed depths for this scanner at $n = 1$ to $n = 500$. According to Eq. 11, The iso-disparity surfaces (Z_n) were plotted by the Epipolar Geometry Toolbox, as shown in Fig. 6 [28]. According to Eq. 16, the camera linear movement Δb to create new surfaces (Z_n^{new*}) between two consecutive iso-disparity surfaces was calculated by 0.04 mm. Figure 7 shows the iso-disparity surfaces with baselines of 250 mm and 250.04 mm, simultaneously. It can be seen that the new iso-disparity surfaces appear between the two previous surfaces. In the second experiment, a laser angle (α) of 11°, camera convergence angle (θ) of 0°, and baseline (b) of 127 mm were selected. Figure 8 shows the reconstructed depths for this scanner at $n = 1$ to $n = 100$. According to Eq. 11, the iso-disparity surfaces (Z_n) were

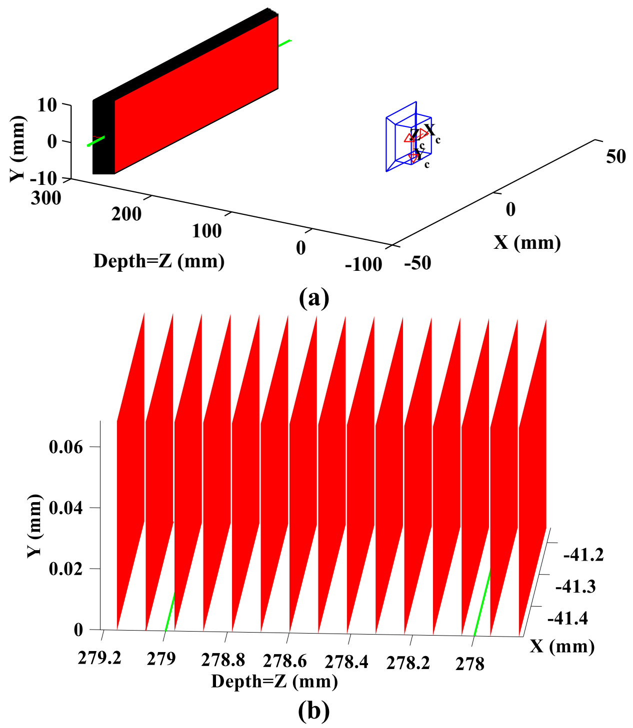


Fig. 6 The iso-disparity surfaces of the laser triangulation scanner with the baseline of 250 mm plotted by the Epipolar Geometry Toolbox. **a** 3D view **b** Zoomed 3D view

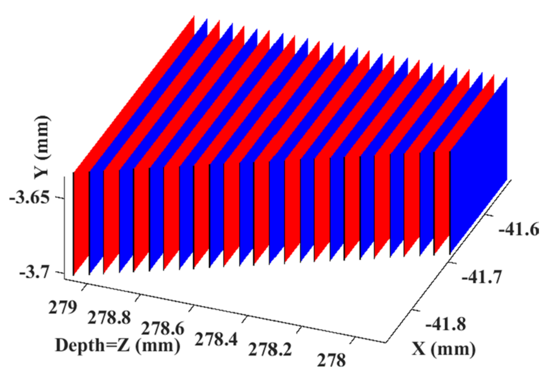


Fig. 7 The iso-disparity surfaces of the laser triangulation scanner with the baseline of 250 mm (the red color) and 250.04 mm (the blue color) plotted by the Epipolar Geometry Toolbox (color figure online)

plotted by the Epipolar Geometry Toolbox, as shown in Fig. 8.

According to Eq. 20, the LRM ($\Delta\alpha$) to create new surfaces ($Z_n^{\text{new}**}$) between two consecutive iso-disparity surfaces was calculated by 0.01°. Figure 9 shows the iso-disparity surfaces with laser angles of 11° and 11.01°, simultaneously. It can be seen that the new iso-disparity surfaces appear between the two previous surfaces. As shown in Figs. 7 and 9, the distance between two sequential iso-disparity surfaces

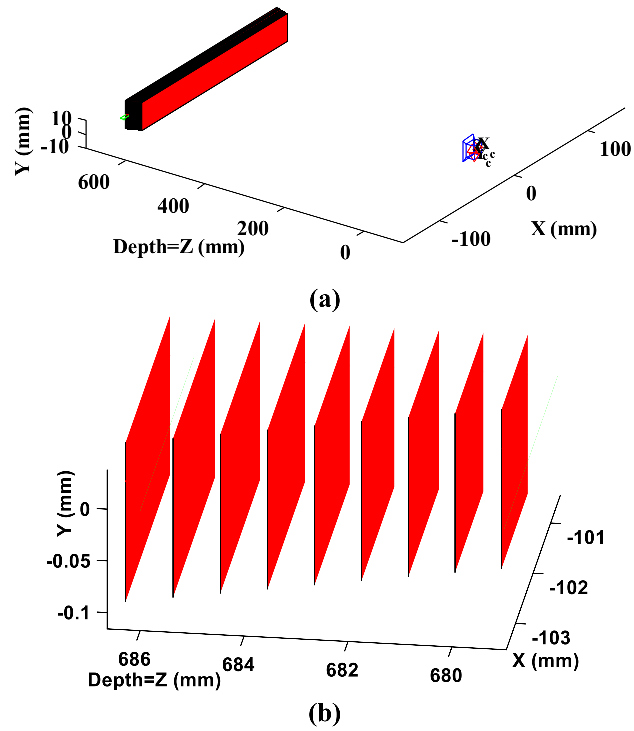


Fig. 8 The iso-disparity surfaces of the laser triangulation scanner with the laser angle of 11° plotted by the Epipolar Geometry Toolbox. **a** 3D view **b** Zoomed 3D view

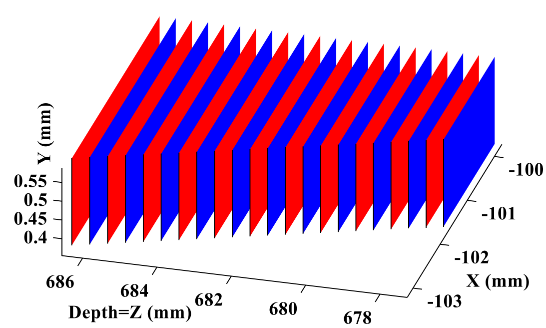


Fig. 9 The iso-disparity surfaces of the laser triangulation scanner with the laser angle of 11° (the red color) and 11.01° (the blue color) plotted by the Epipolar Geometry Toolbox (color figure online)

is reduced by half compared to the scanners shown in Figs. 6 and 8.

It means increasing 50% the reconstruction accuracy measurement. It should be noted that creating a laser scanner with the iso-disparity surfaces shown in Figs. 7 and 9 is impossible; because these figures show the iso-disparity surfaces of two scanners with different setups simultaneously. Accordingly, the mean value of the reconstructed depth from the two scanners with the baselines of 250 mm and 250.04 mm is suggested for the CLM technique in this paper to improve the measurement accuracy. Also, the mean value of the reconstructed depths at the laser angles of 11° and 11.01° is suggested as the final depth according to the LRM technique.

5 The physical experiment setup

Using high-resolution (HR) cameras does not entirely satisfy some users due to the expensive cost of these systems. As a result, a novel method for increasing the 3D measurement accuracy of laser triangulation systems using (LR) and inexpensive cameras is presented in this work. In the following, physical experiments were investigated to validate the mathematical model of laser triangulation disparity and show the proposed method's practicality. The experimental setup of our proposed scanner with the ability of camera linear movement (CLM) and laser rotational movement (LRM) is shown in Fig. 10.

The main components of our scanner were a camera, a laser, and two micrometer tables as described in the following.

5.1 The camera

This paper used a CMOS camera (DFK23GM021, GigE Company, Germany) with a GigE interface that is the perfect solution for industrial automation, quality assurance, security, surveillance, and medical applications. The camera resolution is 1280 × 960 pixels with a sensor size of 1/3" and a pixel size of 3.75 µm. The camera includes a C- to CS-mount adapter, making it compatible with C and CS-mount lenses. Thus, the C-mount lens (M1614-MP2, Computer Company, Japan) with a focal length of 16 mm was used for the physical experiments.

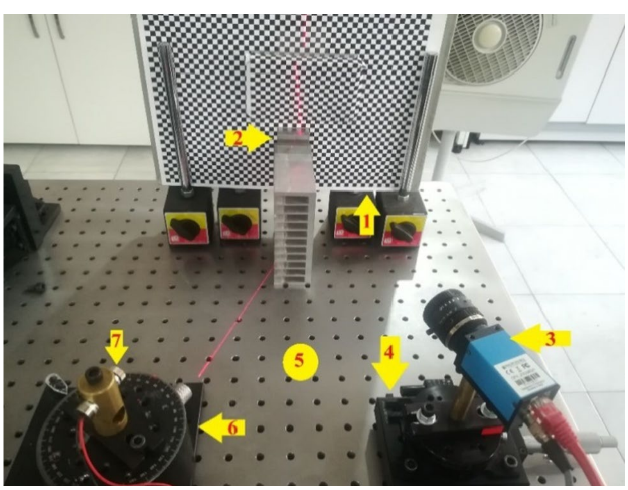


Fig. 10 The experiment setup (1) Calibration plane (2) Workpiece (3) Camera (4) Rotary and translation table (5) Optical breadboard (6) Rotary table (7) Laser

5.2 The laser

A laser line pattern (UHL5-10G-635–90, Worlds star Tech. Company, Canada) with a Gaussian line profile was used. The fan angle and wavelength are 90° and 635 nm, respectively. The laser class is II that is a safety model of Gaussian high-intensity lasers.

5.3 The micrometers tables

A rotary and translation table (Polaritech. Company, Iran) was used to move the camera, as shown in Fig. 11. The maximum linear movement of this table is 25 mm, and the precision of linear motion is 0.01 mm. Along with this linear movement, the maximum rotation of ± 60° with an accuracy of 0.01° is possible. A 360° rotary table with a fine rotation of 0.01° was used to control the laser movement.

6 The physical experiment results

In this study, two chessboard calibration planes containing the squares with an area of 25 mm^2 were used to calibrate the camera and laser simultaneously. The first plane was located just behind the object and was perpendicular to the Z axis. The second was attached on the main calibration plane with an arbitrary angle β as shown in Fig. 12.

By moving the calibration plane to 20 different positions, the spatial measurement space was calibrated by the Matlab Camera Calibration Toolbox [29]. By determining the calibration intrinsic and extrinsic parameters, the reprojection error between the points of all calibration planes and their images is shown in Fig. 13. The reprojection errors were 0.207 and 0.175 pixels along the x- and y-axis, respectively, and it is equivalent to 0.015 mm for

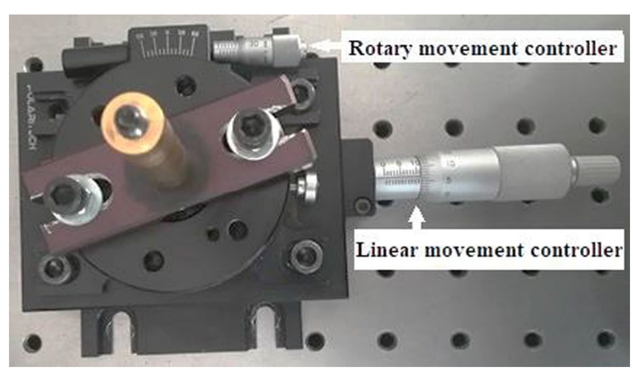


Fig. 11 The rotary and translation table for controlling the camera movement

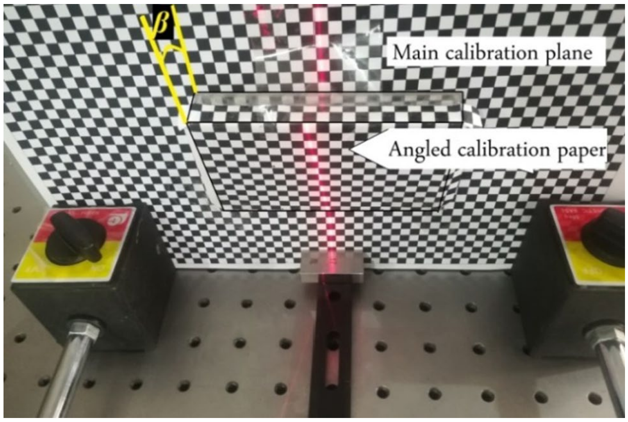


Fig. 12 The main and angle calibration planes with an arbitrary β angle between them

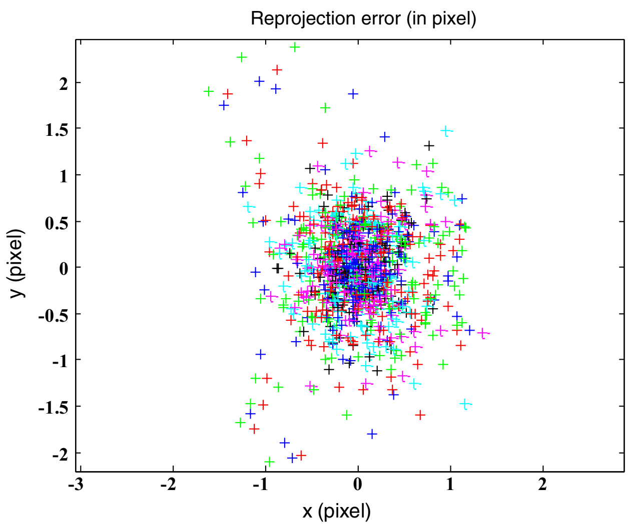


Fig. 13 The reprojection error of the calibration algorithm (each color shows one of the calibration plane points reprojection error) (color figure online)

an image resolution of 1280×960 pixels in our selected working distance. The depth reconstruction of three gage blocks of 70, 24 and17 mm (MW700-02, Moore & Wright with the accuracy of ± 1 μm) was conducted with the CLM and LRM for a physical experiment. The distance of gage blocks to the camera was 270 mm. According to the laser angle and baseline, this distance was selected in such a way that objects were placed in the camera field of view.

The main calibration plane was used for the triangulation algorithm, and the second plane was used for the laser angle measurement. The intersection of the laser plane with each of the two calibrations planes is a line. As shown in Fig. 14, three points located on the two intersection lines were used in this paper to find the laser plane formula based on Eq. 6. Then, one can measure the object point P (see Fig. 1).

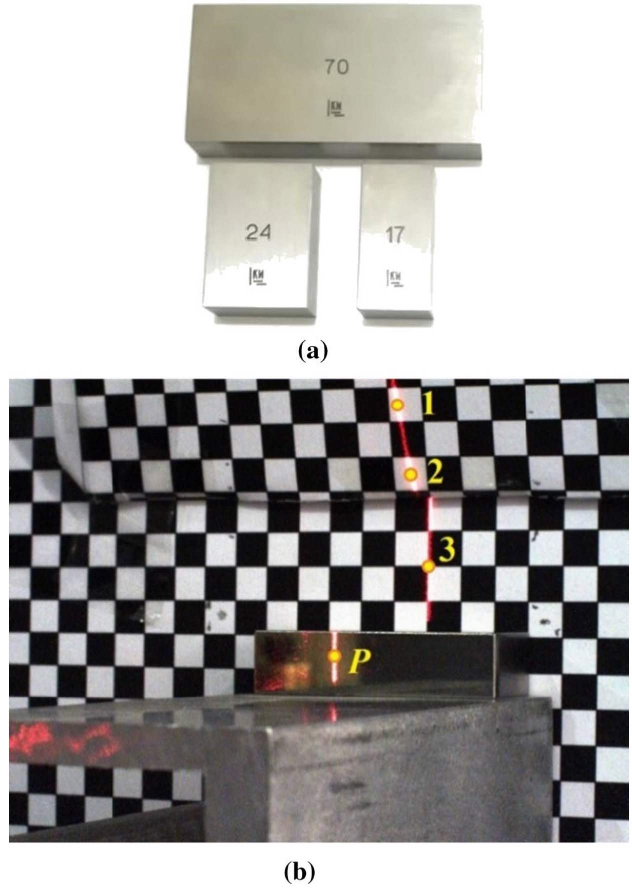


Fig. 14 **a** The gage blocks used for the physical experiment. **b** The gage blocks image from the laser triangulation scanner

6.1 The CLM physical experiment results

According to the synthetic experiment setup, an image was taken from the gage blocks at the laser angle of 20°, camera convergence angle of 30°, and baseline of 250 mm. Then, the depth reconstructions of the blocks were calculated using the camera intrinsic and extrinsic parameters and the calibration algorithm explained before. According to Eq. 17, the camera liner movement of $\Delta b = 0.04$ mm was done by the translation table (Fig. 11). After that, the measurement process was repeated for the new baseline of 250.04 mm. To ensure the reliability of the measurement, the abovementioned algorithms were repeated three times. Finally, the mean value of the measurements for the 250 mm and 250.04 mm baselines was considered as the improved depth.

Consequently, two other physical experiment setups with arbitrary baselines of 350 mm and 450 mm were repeated. According to Eq. 17, the CLM (Δb) of 0.06 and 0.08 mm were calculated for 350 and 450 mm baselines, respectively. Table 1 shows the reconstruction error percentage and the accuracy improvement percentage of a physical experiment using:

Table 1 The CLM physical experiment results

The baseline (mm)	The gage blocks (mm)	Calculated dimension (mm)	Reconstruction error percentage	Final calculated dimension (mm)	Final error percentage	Accuracy improvement percentage
250	17	16.07	5.47	16.36	3.76	31.18
	24	23.38	2.58	23.57	1.79	30.65
	70	70.91	1.30	70.62	0.89	31.87
350	17	16.09	5.35	16.39	3.59	32.97
	24	24.82	3.42	24.55	2.29	32.93
	70	69.21	1.13	69.46	0.77	31.65
450	17	17.89	5.24	17.59	3.47	33.71
	24	23.33	2.79	23.56	1.83	34.33
	70	69.44	0.80	69.63	0.53	33.93

$$\text{Final dimension} = \frac{\text{dimension at } (b) + \text{dimension at } (b + \Delta b)}{2}$$

$$\text{Accuracy improvement percentage} = \frac{|\text{final error percentage} - \text{error percentage at } (b)|}{\text{error percentage at } (b)} \times 100. \tag{21}$$

6.2 The LRM physical experiment results

According to the synthetic experiment setup, an image was taken from the gage blocks at the laser angle (α) of 11°, camera convergence angle (θ) of 0°, and baseline (b) of 127 mm were selected. Then, the depth reconstructions of the blocks were calculated using the camera intrinsic and extrinsic parameters and the calibration algorithm explained before. According to Eq. 20, the laser rotational movement (RLM) of $\Delta\alpha = 0.01°$ was done by the rotary table (Fig. 10). After that, the measurement process was repeated for the new laser angle of 11.01°. Finally, the mean value of the measurements for the laser angles of 11° and 11.01° was considered as the improved depth. Consequently, two other physical experiments setups with the laser angles of 13° and 15° were repeated. According to Eq. 20, the LRM ($\Delta\alpha$) of 0.01° was calculated for both laser angles of 13° and 15°. It should be noted that the laser plane did not intersect the gage blocks if we selected different laser angles with different $\Delta\alpha$. Table 2 shows the reconstruction error percentage and the accuracy improvement percentage of a physical experiment using:

$$\text{Final dimension} = \frac{\text{dimension at } (\alpha) + \text{dimension at } (\alpha + \Delta\alpha)}{2}$$

$$\text{Accuracy improvement percentage} = \frac{|\text{final error percentage} - \text{error percentage at } (\alpha)|}{\text{error percentage at } (\alpha)} \times 100. \tag{22}$$

Table 2 The LRM physical experiment results

The laser angle	The gage blocks (mm)	Calculated dimension (mm)	Reconstruction error percentage	Final calculated dimension (mm)	Final error percentage	Accuracy improvement percentage
11°	17	17.37	2.18	17.27	1.59	27.03
	24	23.44	2.33	23.60	1.67	28.57
	70	70.88	1.26	70.62	0.89	29.55
13°	17	16.12	5.18	16.39	3.59	30.68
	24	23.32	2.83	23.52	2.00	29.41
	70	69.39	0.87	69.57	0.61	29.51
15°	17	16.28	4.24	16.49	3.00	29.17
	24	24.81	3.37	24.58	2.42	28.40
	70	70.91	1.30	70.66	0.94	27.47

6.3 The effect of object's position and alignment on the measurement accuracy

In this paper, we studied the quantization error according to the iso-disparity surfaces. According to Eq. 11, the position and arrangement of iso-disparity surfaces did not depend on the object's position. In the Scheimpflug analysis, the image quality is studied when the object plane of an optical setup is not parallel to the image plane [30, 31]. It should be noted that the position and the alignment of the object affect the measurement accuracy. For this, the following example is described. Suppose the edge of the object to be measured is placed parallel to the iso-disparity surfaces, as shown in Fig. 15.

In this case, the dimension of the object points located on an iso-disparity surface can be calculated without the quantization error, as described in chapter 3. Now, assume an object placed parallel to the iso-disparity surfaces but precisely in the middle of two sequential iso-disparity surfaces. Thus, the maximum quantization error ($R/2$) occurs. Finally, suppose an object that is not parallel to the iso-disparity surfaces, as shown in Fig. 16. According to Fig. 16, the quantization error varies between 0 and $R/2$ depending on the object points' position. It should be noted that we reduced the distance between each two sequential iso-disparity surfaces by half using the CLM or LRM approaches. Thus, the quantization error varies between 0 and R/4. According to the physical experiment setup with $b=250$ mm, $\alpha=20°$, and $\theta=30°$, the gage blocks positioned between $n=49$ ($Z_{n=}$269.95 mm) and $n=50$ ($Z_{n+1}=270.04$ mm) according to Eq. 11. So, the depth resolution $R=Z_{n+1}$-Z_n is 0.09 mm, and the maximum quantization error is R/4 = 0.02 mm. Based on the results presented in Tables 1 and 2, the mean value of our scanner accuracy was 0.8 mm. Thus, the maximum quantization error is 2.5% which can be ignored.

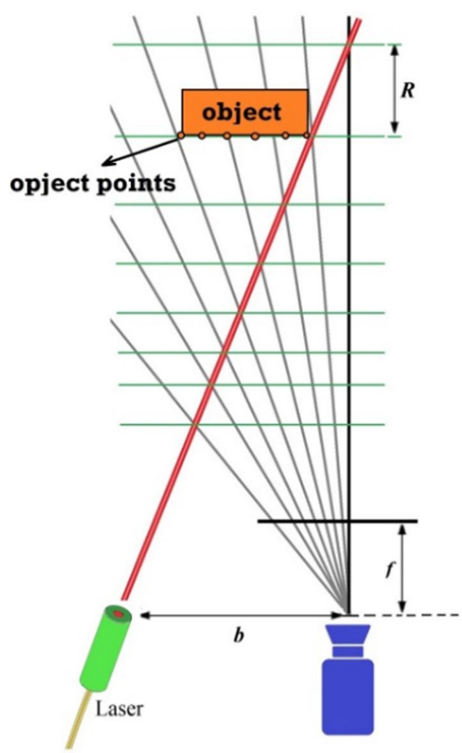


Fig. 15 The object points located on an iso-disparity surface

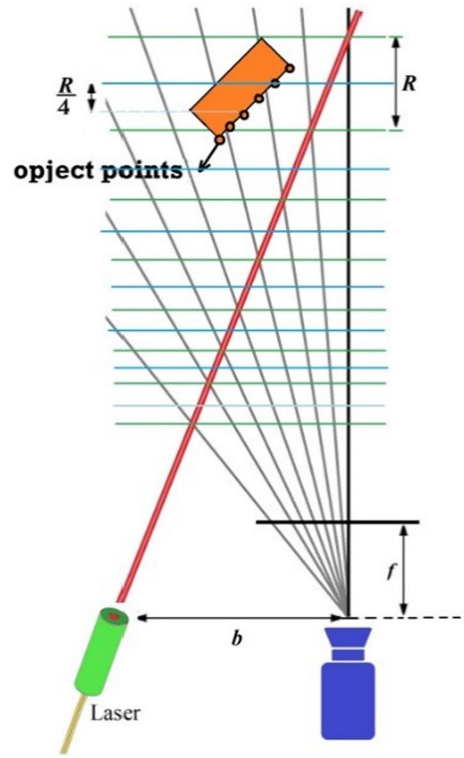


Fig. 16 The iso-disparity surfaces for the first setup (the green color) and after the CLM or LRM approaches (the blue color) surfaces

7 Conclusions

The accuracy assessment of the laser triangulation scanner by adjusting the relative position of the camera and laser is presented in this study. According to the proposed mathematical model, the laser angle (α), the baseline (b), and the camera convergence angle (θ) are the scanner setup parameters that can be changed to rearrange the iso-disparity surfaces to improve the measurement accuracy. In this study, the baseline (b) and the laser angle (α) were selected to investigate the effect of camera linear movement (CLM) and laser rotational movement (LRM) on the measurement accuracy improvement of the laser triangulation scanners. The advantages and findings of our proposed scanner can be summarized as:

1. According to the sensor resolution limitation, the image coordinate of each scene point is extracted by a pixel center coordinate leads to the quantization error. This error causes an arrangement of parallel iso-disparity surfaces in front of the laser scanners. The depth resolution is the distance between two sequential iso-disparity surfaces and can be considered for investigating the quantization error.
2. For 50% improving the measurement accuracy or reducing the quantization error, it will be needed to create a surface between each two consecutive iso-disparity surfaces.
3. This paper proposes the camera linear movement and laser rotational movement to reduce the iso-disparity surface intervals. The CLM and LRM were investigated by changing the baseline and laser angle, respectively.
4. The mathematical models were extracted for CLM and LRM techniques. The synthetic experiments showed that arranging the new iso-disparity surfaces is possible in the middle of old surfaces.
5. The physical experiment results showed the mean accuracy improvement percentage of 32.58% and 28.87% for the CLM and LRM techniques, respectively. The resolution limitation of rotary and translation tables was the most important reason to improve the measurement accuracy by less than 50%.
6. In our physical experiment setups, the camera liner movement (Δb) was bigger than the translation table resolution (0.01 mm), but the laser rotational movement ($\Delta \alpha$) was equal to the rotary table resolution (0.01°). Thus, the measurement accuracy improvement of the CLM method was higher than the LRM method according to the physical experiment results. It should be noted that improving the measurement accuracy near to 50% can be achievable by both LRM and CLM techniques using the HR rotary and translation tables.
7. Improving the measurement accuracy by more than 50% is possible using the CLM or LRM method by more camera linear or laser rotational movements, but it will increase the measurement time.
8. This study presents sub-pixel resolution algorithms to improve the measurement accuracy of calculating the 3D coordinates of object points. By moving the object crossing the laser line, the 3D coordinates of a set of points on the external surface of objects known as the point cloud can be produced. Then, the point cloud will be ready to convert to the 3D model.

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

References

1. Silva RHGOE, Galeazzi D, Schwedersky MB, Mendonça FK, Bonamigo AV, Marques C (2021) An adaptive orbital system based on laser vision sensor for pipeline GMAW welding. J Braz Soc Mech Sci Eng 43:1–17
2. Xie Z, Zhang Z, Jin M (2006) Development of a multi-view laser scanning sensor for reverse engineering. Meas Sci Technol 17:2319
3. Hernandez ID, Vaz MA, Cyrino JC, Alvarez NM (2019) Three-dimensional image-based approach for imperfect structures surface modeling. J Braz Soc Mech Sci Eng 41:1–21
4. Leclet-Groux D, Caron G, Mouaddib E, Anghour A (2013) A serious game for 3D cultural heritage. In: 2013 Digital Heritage International Congress (DigitalHeritage), 409–412.
5. Deak G, Curran K, Condell J (2012) A survey of active and passive indoor localisation systems. Comput Commun 35:1939–1954
6. He Z, Kang L, Zhao X, Zhang S, Tan J (2020) Robust laser stripe extraction for 3D measurement of complex objects. Measur Sci Technol.
7. Vukašinović N, Bračun D, Možina J, Duhovnik J (2012) A new method for defining the measurement-uncertainty model of CNC laser-triangulation scanner. Int J Adv Manuf Technol 58:1097–1104
8. Feng HY, Liu Y, Xi F (2001) Analysis of digitizing errors of a laser scanning system. Precis Eng 25:185–191
9. Xi F, Liu Y, Feng HY (2001) Error compensation for three-dimensional line laser scanning data. Int J Adv Manuf Technol 18:211–216
10. Khalili K, Razavi SA, Karimzadgan D (2005) High resolution measurements using a low resolution system. Measur Sci Rev 5:56–59
11. Sayyedbarzani SA, Emam SM (2020) Evaluation of the quantization error in convergence stereo cameras. J Opt Technol 87:495–500
12. Sayyedbarzani SA, Emam SM (2021) 3D reconstruction accuracy improvement of a stereo vision system by changing the convergence angle of the cameras. J Opt Technol 88:574–578
13. Alasseur C, Constantinides AG, Husson L (2003) Colour quantisation through dithering techniques. In: Proceedings 2003 International Conference on Image Processing, I-469
14. S. Mukherjee, G.M. Su, I. Cheng, Adaptive dithering using Curved Markov-Gaussian noise in the quantized domain for mapping SDR to HDR image, In International Conference on Smart Multimedia (2018) 193–203.
15. Hook RN, Fruchter AS (2000) Dithering, sampling and image reconstruction. In Astronomical data analysis software and systems.
16. Ritz M, Langguth F, Scholz M, Goesele M, Stork A (2012) High resolution acquisition of detailed surfaces with lens-shifted structured light. Comput Graph 36:16–27
17. Hattori K, Sato Y (1996) Accurate rangefinder with laser pattern shifting. In: Proceedings of 13th International Conference on Pattern Recognition, 849–853
18. Ben-Ezra M, Zomet A, Nayar SK (2006) Jitter camera: a super-resolution video camera. In: Visual Communications and Image Processing (vol 6077, p. 607704), International Society for Optics and Photonics.
19. Huang X, Qin R, Xiao C, Lu X (2018) Super resolution of laser range data based on image-guided fusion and dense matching. ISPRS J Photogramm Remote Sens 144:105–118
20. Ma L, Li Y, Li J, Wang C, Wang R, Chapman MA (2018) Mobile laser scanned point-clouds for road object detection and extraction: a review. Remote Sens 10:1531

21. Li B, Zhou Y, Zhang Y, Wang A (2020) Depth image super-resolution based on joint sparse coding. Pattern Recogn Lett 130:21–29
22. Du J, He Z, Wang L, Gholipour A, Zhou Z, Chen D, Jia Y (2019) Super-resolution reconstruction of single anisotropic 3D MR images using residual convolutional neural network. Neurocomputing.
23. Kil YJ, Mederos B, Amenta N (2006) Laser scanner super-resolution. Eurograph Symp Point Based Graph, 9–16.
24. Kurka PRG, Delgado JV, Mingoto CR, Rojas OER (2013) Automatic estimation of camera parameters from a solid calibration box. J Braz Soc Mech Sci Eng 35:93–101
25. Abu-Nabah BA, ElSoussi AO, Al Alami AEK (2016) Simple laser vision sensor calibration for surface profiling applications. Opt Lasers Eng 84:51–61
26. Nikolova I, Nikolov A, Zapryanov G (2011) Depth estimation using shifted digital still camera. In: Proceedings of the 12th International Conference on Computer Systems and Technologies, 234–240.
27. Pollefeys M, Sinha S (2004) Iso-disparity surfaces for general stereo configurations, In: European Conference on Computer Vision, 509–520
28. Mariottini GL, Prattichizzo D (2005) The epipolar geometry toolbox: multiple view geometry and visual servoing for MATLAB. IEEE Robotics and Automation Magazine
29. Bouguet JY (2004) Camera Calibration Toolbox for Matlab 2004, URL http://www.vision.caltech.edu/bouguetj/calibdoc.
30. Mei Q, Gao J, Lin H, Chen Y, Yunbo H, Wang W, Chen X (2016) Structure light telecentric stereoscopic vision 3D measurement system based on Scheimpflug condition. Opt Lasers Eng 86:83–91
31. Fasogbon P, Duvieubourg L, Lacaze PA, Macaire L (2015) Intrinsic camera calibration equipped with Scheimpflug optical device. In: Twelfth International Conference on Quality Control by Artificial Vision, 953416.